Software Engineering
Assignment-1
TYPE OF QUESTION: MCQ/MSQ

Number of questions: **10** **Total mark: 10 X 1 = 10**

QUESTION 1:

Which of the following are considered major challenges faced in modern software development projects?

a. Software maintenance costs are very high.
b. Projects often exceed their scheduled timelines.
c. Increasing hardware speed completely eliminates software quality issues.
d. Users expect software systems to be reliable and user-friendly.
e. Software systems are becoming increasingly large and complex.

Correct Answer: **a**. Software maintenance costs are very high.
b. Projects often exceed their scheduled timelines.
d. Users expect software systems to be reliable and user-friendly.
e. Software systems are becoming increasingly large and complex.

Detailed Solution:

The major challenges in modern software development include high maintenance costs, schedule overruns, growing user expectations for reliability and usability, and increasing system size and complexity—all of which make projects harder to manage and deliver. However, faster hardware does not eliminate software quality issues, as poor design, bugs, and security flaws remain independent of processing power.

QUESTION 2:

Which of the following statements are characteristics of structured programming?

a. A program is divided into smaller modules or functions
b. The use of excessive GOTO statements is encouraged
c. Control structures such as sequence, selection, and iteration are used
d. Proper indentation improves program readability
e. Structured programming discourages code documentation

Correct Answer: a. A program is divided into smaller modules or functions
c. Control structures such as sequence, selection, and iteration are used
d. Proper indentation improves program readability

Detailed Solution:

Structured programming promotes modularity (dividing programs into functions), uses disciplined control structures like sequence, selection, and iteration, and encourages proper indentation for readability—all of which improve clarity and maintainability. It explicitly discourages excessive GOTO statements and does not discourage documentation; in fact, clear documentation is often encouraged alongside self-explanatory code.

QUESTION 3:

Which of the following statements support the idea that "The more a software product is maintained, the less maintainable it becomes"?

a. Frequent changes can increase software complexity
b. Poorly documented modifications reduce code understandability
c. Continuous maintenance always improves software structure
d. Repeated patches may introduce inconsistencies in design
e. Lack of proper refactoring during maintenance can degrade code quality

Correct Answer: a. Frequent changes can increase software complexity
b. Poorly documented modifications reduce code understandability
d. Repeated patches may introduce inconsistencies in design
e. Lack of proper refactoring during maintenance can degrade code quality

Detailed Solution:

Frequent changes, poor documentation, repeated patches, and lack of refactoring gradually degrade a software system's structure, making it harder to understand and modify over time—this is known as software entropy. The statement that continuous maintenance always improves structure is false, because without deliberate effort, maintenance often introduces new flaws and inconsistencies.

QUESTION 4:

Which of the following are generally categorized as software service projects?

a. Software maintenance
b. Development of a commercial off-the-shelf software product
c. Software customization for a client
d. Outsourced software development
e. Software testing and support services

Correct Answer: a. Software maintenance
c. Software customization for a client
d. Outsourced software development
e. Software testing and support services

Detailed Solution:

Software service projects involve providing ongoing support, custom solutions, or contract-based work for specific clients, such as maintenance, customization, outsourced development, and testing services. In contrast, developing a commercial off-the-shelf (COTS) product is a product-development effort, not a service project, as it is built for mass sale rather than for a particular client.

QUESTION 5:

Based on the distinction between jobs, projects, and exploration activities, which of the following statements are correct?

a. Jobs usually involve repetitive and well-understood tasks
b. Exploration activities generally have a high degree of uncertainty in outcomes
c. Projects typically contain both routine activities and challenging tasks
d. Projects always have completely predictable outcomes
e. Exploration activity may encompass searching for new scientific discoveries

Correct Answer: a. Jobs usually involve repetitive and well-understood tasks
b. Exploration activities generally have a high degree of uncertainty in outcomes
c. Projects typically contain both routine activities and challenging tasks
e. Exploration activity may encompass searching for new scientific discoveries

Detailed Solution:

Jobs are routine and well-understood, exploration involves high uncertainty (e.g., scientific discovery), and projects mix routine work with novel challenges—but they never guarantee completely predictable outcomes. Therefore, the claim that projects always have predictable results is false, while the others accurately describe these activity types.

QUESTION 6:

Which of the following statements regarding types of software are correct?

a. Packaged software is prewritten software available for purchase
b. Custom software is usually developed according to specific user requirements
c. Horizontal market software is designed to satisfy the needs of many companies
d. Vertical market software is intended for a particular industry domain
e. Packaged software is always developed for only one specific customer

Correct Answer: a. Packaged software is prewritten software available for purchase
b. Custom software is usually developed according to specific user requirements
c. Horizontal market software is designed to satisfy the needs of many companies
d. Vertical market software is intended for a particular industry domain

Detailed Solution:

Packaged software is prewritten for mass purchase, custom software is built for specific user needs, horizontal software serves many companies (e.g., payroll), and vertical software targets a specific industry (e.g., hospital management). However, packaged software is not developed for just one customer—that describes custom software, making option e incorrect.

QUESTION 7:

Why should software developers focus on writing readable and well-structured code?

a. Readable code makes debugging easier
b. Readable code helps other developers understand the program more easily
c. Readable code always guarantees minimum memory usage
d. Readable code simplifies software maintenance and modification
e. Readable code facilitates effective testing of the software

Correct Answer: a. Readable code makes debugging easier
b. Readable code helps other developers understand the program more easily
d. Readable code simplifies software maintenance and modification
e. Readable code facilitates effective testing of the software

Detailed Solution:

Readable code improves debugging, collaboration, maintenance, and testing by making the program's logic clear and easy to follow. However, it does not guarantee minimum memory usage—that depends on algorithmic and data-structure choices, not readability alone.

QUESTION 8:

Which of the following projects can be classified as software service-type projects?

a. Providing long-term support and maintenance for a hospital management system
b. Customizing an existing payroll software package for a specific company
c. Developing a banking application for a particular client organization
d. Developing a generic photo-editing software product for worldwide commercial sale
e. Migrating and integrating legacy software systems for a client

Correct Answer: a. Providing long-term support and maintenance for a hospital management system
b. Customizing an existing payroll software package for a specific company
c. Developing a banking application for a particular client organization
e. Migrating and integrating legacy software systems for a client

Detailed Solution:

Service-type projects involve providing support, customization, or bespoke development for specific clients—such as maintenance, customizing payroll, building a banking application for one client, or migrating legacy systems. Developing a generic photo-editing product for worldwide sale is a product-development effort (COTS), not a service project.

QUESTION 9:

Which of the following issues are commonly associated with the build-and-fix model of software development?

a. Coding begins without complete requirement analysis
b. Little effort is made toward proper software design
c. Code is systematically documented during development
d. The model guarantees highly maintainable software
e. Testing is usually unsystematic and performed late
f. The resulting software may suffer from poor quality and maintainability

Correct Answer: a. Coding begins without complete requirement analysis
b. Little effort is made toward proper software design
e. Testing is usually unsystematic and performed late
f. The resulting software may suffer from poor quality and maintainability

Detailed Solution:

The build-and-fix model starts coding without proper analysis or design, uses unsystematic testing done late, and typically produces low-quality, hard-to-maintain software. It does not encourage systematic documentation and certainly does not guarantee maintainability.

QUESTION 10:

Which of the following situations illustrate the use of the principle of abstraction in software development?

a. Representing the logic of a program using pseudocode before implementation
b. Dividing a large software system into smaller modules
c. Designing control flow structure before writing code
d. Ignoring unnecessary implementation details during system design
e. Testing whether a program produces correct outputs for various inputs

Correct Answer: a. Representing the logic of a program using pseudocode before implementation
c. Designing control flow structure before writing code
d. Ignoring unnecessary implementation details during system design

Detailed Solution:

Abstraction means focusing on essential ideas while hiding irrelevant details—using pseudocode, designing control flow at a high level, and ignoring implementation specifics during design all exemplify this. Testing outputs (option e) is about validation, not abstraction, and dividing into modules (option b) is decomposition, though closely related.

******THE END******